Inter-Research Science Publisher

Endangered Species Research
Volume 56:159–172

Published April 4, 2025
https://doi.org/10.3354/esr01390

OPEN ACCESS

Unexpected population genetic structure in Bayou darter *Nothonotus rubrus* in the geomorphically dynamic Bayou Pierre, Mississippi, USA

Loren W. Stearman*, Brian R. Kreiser, Jacob F. Schaefer

School of Biological, Environmental, and Earth Sciences, The University of Southern Mississippi, Hattiesburg, MS 39406, USA

ABSTRACT: Habitat fragmentation reduces gene flow and population resiliency. Fragmentation can contribute to localized extinctions, and for small-range taxa, range-wide extinction. Accelerated erosion in streams and rivers may fragment habitats for habitat-specialist species by greatly altering river mainstems, pushing many taxa into isolated tributaries. In this paper, we examine patterns of genetic structuring in the federally threatened and endemic Bayou darter *Nothonotus rubrus* to assess whether patterns of erosion are causing fragmentation of populations. Patterns of heterozygosity support the hypothesis of recent colonization of upper White Oak Creek. Analysis using STRUCTURE software yielded unexpected results: a population isolated above an erosional feature, a population widespread throughout the watershed, and evidence of introgression between 2 of 3 genetic clusters. Multivariate analyses support STRUCTURE in suggesting evidence for 3 genetically distinct populations. Effective population size analysis found estimates of abundance to be spatially concordant with recent field studies. Our results demonstrate unexpected population structuring in *N. rubrus*, potentially due to populations becoming separated by features related to in-channel erosion. The amount of local adaptation related to this differentiation is unknown; however, conservation efforts and efforts to augment existing populations should take into account genetic variation and potential local adaptation. We recommend next steps for further exploring genetic variation and potential conservation strategies for this species.

KEY WORDS: Single-nucleotide polymorphisms · SNPs · Population fragmentation · Effective population size · Channel evolution · Erosion · Knickpoints

1. INTRODUCTION

Habitat fragmentation is a pervasive feature of many modern riverscapes, and its effects on resiliency and connectivity of individual populations are often deleterious. Human activities, such as infrastructure development, fragmentation of rivers by dams, and direct habitat alterations, degrade habitats and block migration routes for many organisms. These alterations limit dispersal and patch connectivity for less-mobile species (Fuller et al. 2015). The implications for migratory species can be clear, as lack of access to habitats required for a particular life stage can disrupt life cycles and extirpate populations (e.g. migratory lampreys, pelagic egg drift minnows, Perkin & Gido 2011, Keefer et al. 2013, Clemens & Wang 2021). But even for less-mobile species, disruption of dispersal, rescue effects, and source–sink dynamics can elevate local population extinction risk and alter local community formation (Perkin & Gido 2012, Stearman & Lynch 2013, Brauer & Beheregaray 2020). Fragmentation may result in lower rates of gene flow, reducing the portfolio effect, increasing inbreeding depression, and weakening population resiliency against disease or environmental variation for populations which persist in fragmented habitats (Carlson & Satterthwaite 2011, Sterling et al. 2012, Coleman et al. 2018). The effects of fragmentation may be

*Corresponding author: loren.stearman@usm.edu

© The authors 2025

Publisher: Inter-Research · www.int-res.com

particularly pronounced for aquatic organisms. Populations of these organisms are already more fragmented than in terrestrial landscapes due to their reliance on rivers and streams for dispersal corridors and difficulty dispersing around barriers (Fagan 2002, Faulks et al. 2011, Altermatt 2013). Small-range species are especially at risk, where even minor habitat loss or fragmentation can drive populations to extinction or near-extinction (e.g. desert pupfishes, Carson et al. 2014, Valdés González et al. 2020).

Channel evolution in alluvial streams and rivers due to accelerated fluvial erosion can have numerous complicated impacts on aquatic habitats and organisms, including habitat fragmentation. Stream channels often evolve through characteristic morphological phases as they respond to sudden changes in hydrological, sedimentary, or wood regimes (Simon & Rinaldi 2006, Van Dyke 2013, Cluer & Thorne 2014; our Fig. 1). Stream channel response to an increase in net stream power (i.e. from downstream straightening or mining) typically involves incision and channel widening, causing disconnection from floodplain habitats (Wohl 2015). Channel straightening associated with incision can also lead to increased distances between riffle habitats as the pools between them lengthen, and an increased export of materials can result in lower numbers and quality of riffle habitats (Slack et al. 2004, Lofthouse & Robert 2008). At broader scales, these same processes can result in unstable sediment supply and unstable fluvial land-

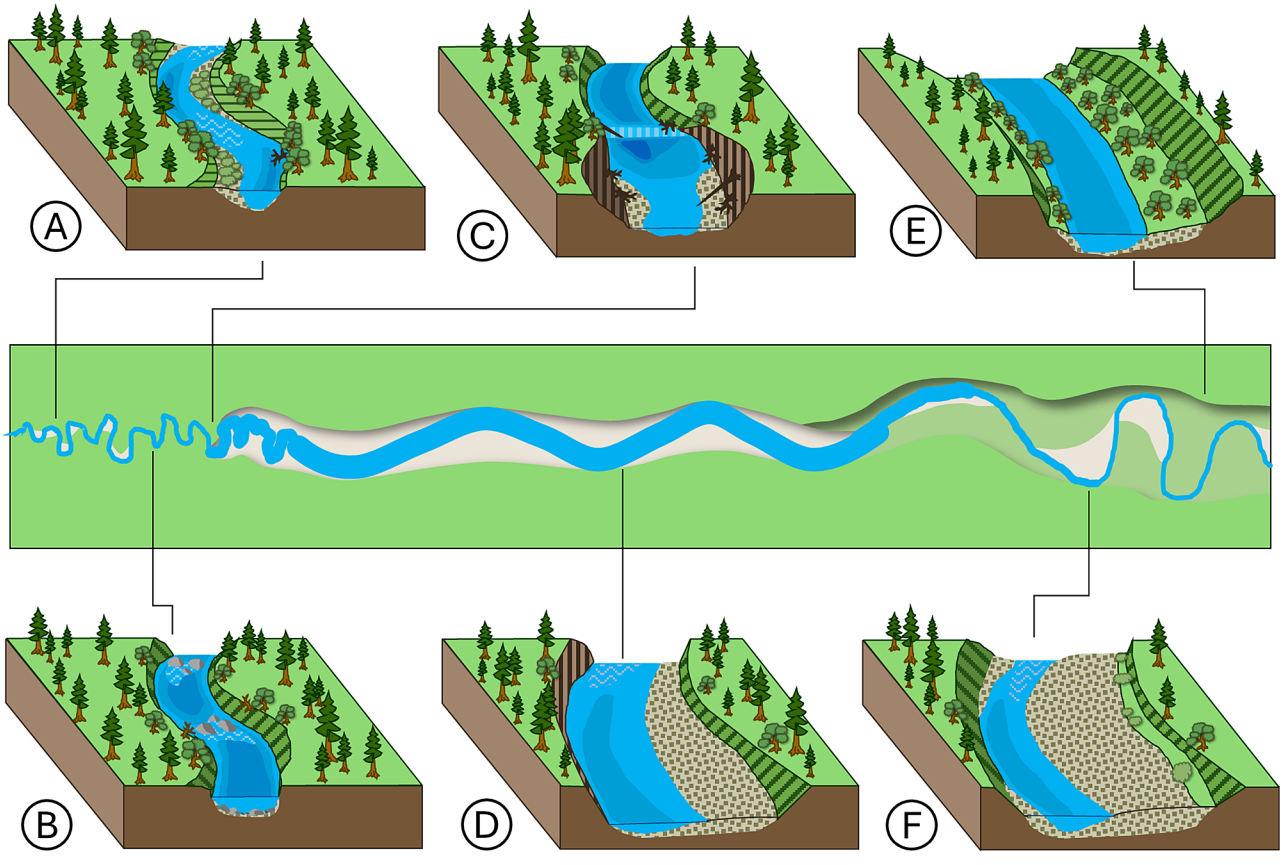

Fig. 1. Model of typical channel evolution processes in Bayou Pierre, where net negative sediment budgets have caused considerable changes to aquatic habitats. Flow is from left to right, and net hydraulic discharge is assumed to be constant throughout the figure. Note changes in sinuosity, width:depth ratios, and entrenchment or inset floodplain conditions. (A) Pre-erosion conditions; (B) incipient erosion with modest incision, leading to incised channels and often coarser substrates; (C) catastrophic bank failure, further incision, channel widening, and chaotic sinuosity with knickpoint passage; (D) continued channel widening, incision, and sinuosity loss, with large sediment bars and high rates of sediment transport; (E,F) development of string and bead morphology (Stanford et al. 1996) with high rates of channel migration and alternating reaches with little sediment retention (E) or exceptionally large, unstable bar deposits (F). This model is based on channel evolution models from Simon & Rinaldi (2006), Cluer & Thorne (2014), and data from Stearman & Schaefer (2022, 2023) and Stearman & Heitmuller (2024). Pre-erosion conditions (A) are currently restricted to the extreme headwaters. The majority of the contemporary distribution of *Nothonotus rubrus* is in reaches experiencing conditions in (D). Downstream reaches where our sampling and Hubbell et al. (2021) failed to capture *N. rubrus* are largely in (E,F)

forms, creating patchy and unpredictable in-channel habitats (Gran & Czuba 2017, Stearman & Schaefer 2022). These changes often progress upstream away from a point of disturbance, with a leading edge forming an erosional crest and a knickpoint — a characteristic break in longitudinal profile which is often associated with dramatic bank failure, channel widening, and chaotic sinuosity (Simon & Rinaldi 2006; our Fig. 1). Upstream migration of the erosional crest may functionally trap pre-eroded habitats in the tributary network, with the trunk and larger tributaries hosting greatly altered habitats (Stearman 2023). The ecological impact of channel evolution can be as marked as the geomorphic changes themselves. At local scales, channel evolution often results in the loss of riffle and pool specialist species (Shields et al. 1994, Ross et al. 2001, Tipton et al. 2004). Species that are successful in eroded channels often have traits associated with increased mobility, ecological generalism, and rapid population turnover (Stearman & Schaefer 2022). At watershed scales, this can result in lost diversity, increased community homogenization, and changes in metapopulation and metacommunity dynamics (Stearman & Schaefer 2022, 2023). In extreme cases, channelization due to erosion can result in fragmentation of populations and marked reductions in gene flow for headwater species (Sterling et al. 2012, Stearman & Schaefer 2024).

The federally threatened Bayou darter *Nothonotus rubrus*, originally described by Raney & Suttkus (1966), is a small benthic fish endemic to the Bayou Pierre, Mississippi. *N. rubrus* has been of conservation concern due to accelerated fluvial erosion in its watershed for nearly as long as it has been studied (Suttkus & Clemmer 1977, Matthews 1978). *N. rubrus* are riffle specialists, preferring swift shallow water over stable coarse pebble and cobble substrates (Ross et al. 1992). Such habitats are uncommon in the sand and sand-gravel bed streams of the Gulf Coastal Plain, and erosional activities within the Bayou Pierre may play a role in their distribution and connectivity (Stearman & Schaefer 2022, 2023, Stearman & Heitmuller 2024). Ross et al. (2001) observed lower densities of *N. rubrus* in riffle habitats in lower and more eroded portions of the Bayou Pierre, including regions of historically abundant collections by Suttkus & Clemmer (1977), suggesting that habitat loss due to accelerated fluvial erosion negatively affects individual populations of this species. Interestingly, Ross et al. (2001) also found densities of *N. rubrus* to be highest in regions of incipient erosion rather than less-disturbed habitats. Ross et al. (2001) attributed this to the coarser substrates and steeper gradients in such stream reaches. Hubbell et al. (2021) found that populations of this species have shifted upstream over a period of several decades, suggesting that they are following a migrating erosional crest.

The upstream migration of *N. rubrus*, coupled with decreased population densities in downstream reaches, poses a potential concern for fragmentation of populations. Although little information is available on adult dispersal through non-riffle habitats, adults of the closely related yellowcheek darter *N. moorei* are notable for a strong avoidance of pool habitats, even in drought conditions, indicating restricted adult mobility and dispersal (Weston et al. 2010). Larval drift has been documented for *N. rubrus*, but survival rates of larvae and juveniles in downstream eroded habitats is unknown (Slack et al. 2004). As populations continue to disperse upstream into the tributary network, they may therefore encounter an increasing barrier to gene flow and decreased metapopulation rescue effects. Little information currently exists on how erosional patterns are affecting gene flow and genetic diversity of *N. rubrus*. Slack et al. (2010) documented little genetic diversity in the species, but their analysis was limited to 4 sample localities and they only examined sequences from the mitochondrial control region. In this paper, we improve our understanding of genetic diversity of *N. rubrus* in relation to ongoing fluvial erosion in the Bayou Pierre. We employed fine-scale genetic tools to build a stronger understanding of genetic variation in this species. We characterize geographic patterns of genetic structure and compare effective population size (N_e) and heterozygosity across sites. We predict that populations of *N. rubrus* should exhibit more genetic structuring and patterns consistent with recent colonization, such as lower heterozygosity or N_es, in the upstream portion of their contemporary distribution.

2. MATERIALS AND METHODS

2.1. Study watershed

Bayou Pierre is a 6th-order tributary of the lower Mississippi River, draining 2770 km^2 in southwest Mississippi (Fig. 2). The watershed is predominantly low gradient with rolling hills and has an alluvial geology comprised of fluvial deposits from Miocene to recent, with some localized sandstone outcrops in tributaries and loess deposition near the bluff line (Horton et al. 2017, Stearman & Schaefer 2023). Pre-

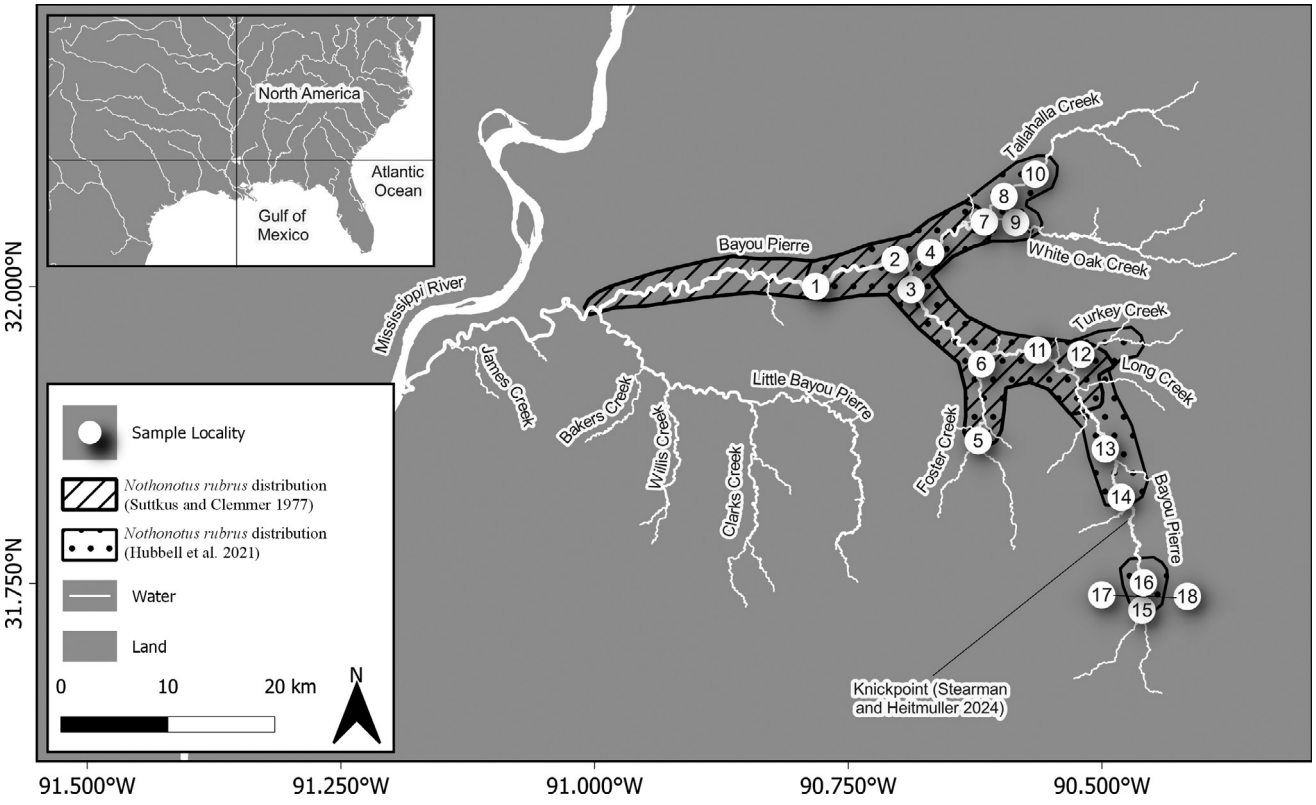

Fig. 2. Collection localities for *Nothonotus rubrus* in Bayou Pierre. Site numbers correspond with Table 1 and with subsequent figures. Distribution from Suttkus & Clemmer's (1977) status update is shown in diagonal crosshatching, while distribution from Hubbell et al.'s (2021) status update is shown with dotted polygon. The location of the knickpoint on the mainstem (upstream extent of erosional activity) is indicated. Note the disjunct population in the upper mainstem Bayou Pierre. Inset: location of Bayou Pierre in southeastern USA. Crosshairs indicate watershed location

Euroamerican settlement, the watershed was likely longleaf pine forest managed by indigenous peoples. Historical deforestation occurred in 2 periods (early Euroamerican settlement, 1830–1860; Piney Woods lumber boom, 1890–1920), followed by extensive reforestation in subsequent decades (Stearman & Heitmuller 2024). There are no large dams and few small impoundments in the watershed. Culverts are restricted to extreme headwaters, and in-channel erosion control structures are absent. Bayou Pierre is undergoing considerable erosion due to both historic (migration of the Mississippi River, Mississippi River channelization) and recent influences (large storm events, Stearman & Heitmuller 2024). These processes have led to greatly altered distribution, stability, and availability of aquatic habitats. Notably, this involves reduced riffle–pool structure and connectivity in the mainstem and larger tributaries (Stearman & Schaefer 2023), and a pronounced knickpoint at the leading edge of erosion throughout the watershed (Fig. 2).

2.2. Tissue acquisition and processing

We sampled tissues from 284 *Nothonotus rubrus* from 19 localities in 2021 in the Bayou Pierre, representing the distribution of recent collections of the species (Table 1, Fig. 2). We collected fish using both seining and electrofishing, often in conjunction with community sampling efforts, with a target sample size of ⩾10 individuals (minimum acceptable of 5) per site. We removed approximately 0.1 g of tissue from fins (typically pelvic fins) and stored tissues in 95% ethanol on ice. Individuals were released alive in their microhabitat of capture. We also included focal sample localities at the upstream edge of known distributions of the species in White Oak, Tallahalla, and Foster Creeks, and the upper Bayou Pierre. Following collection, tissues were transported to the University of Southern Mississippi and stored at −20°C.

We used genotyping-by-sequencing (GBS, Elshire et al. 2011) to obtain single-nucleotide polymorphisms (SNPs) as high-resolution molecular markers

Table 1. Collection localities and number of individuals sequenced for *Nothonotus rubrus*. Effective population size (N_e) region abbreviations are LBP: Lower Bayou Pierre; WOC: White Oak Creek; MBP: Middle Bayou Pierre; UBP: Upper Bayou Pierre; and BPH: Bayou Pierre Headwaters. Individuals from site 4 did not pass quality filters during genotyping and are excluded from the present study; however, raw data are available at our Dryad deposition (see 'Data availability'). N_e: effective population size; N/A: not applicable

Site	Latitude (°N)	Longitude (°W)	Watershed	Tissues genotyped	N_e region
1	32.005	90.784	Bayou Pierre	6	LBP
2	32.022	90.692	Bayou Pierre	9	LBP
3	32.003	90.690	Bayou Pierre	9	LBP
4	32.029	90.679	White Oak Creek	4	LBP
5	31.876	90.623	Foster Creek	8	MBP
6	31.941	90.620	Foster Creek	4	MBP
7	32.060	90.617	White Oak Creek	7	WOC
8	32.082	90.598	Tallahalla Creek	7	WOC
9	32.060	90.586	White Oak Creek	8	WOC
10	32.101	90.567	Tallahalla Creek	3	WOC
11	31.952	90.564	Bayou Pierre	9	MBP
12	31.949	90.522	Turkey Creek	6	UBP
13	31.869	90.497	Bayou Pierre	10	UBP
14	31.828	90.481	Bayou Pierre	5	UBP
15	31.733	90.461	Bayou Pierre	6	BPH
16	31.757	90.459	Bayou Pierre	5	BPH
17	31.745	90.457	Bayou Pierre	4	BPH
18	31.739	90.456	Bayou Pierre	4	BPH

from a total of 144 individuals. We extracted DNA using DNEasy Blood and Tissue Kits (Qiagen). Following extraction, DNA fragmentation and library prep were conducted by Elshire Group. DNA was fragmented using the *Eco*T221 restriction enzyme, and fragments were amplified with PCR. Sample fragments were then sequenced on an Illumina HiSeq platform. At the University of Southern Mississippi, we then demultiplexed sequences and removed combinatorial barcodes using AXE-demux (Murray & Borevitz 2018). Individuals with low overall read counts (<300 000) were then removed from the dataset. We filtered reads to retain only quality scores >30. Raw sequence data were analyzed with the TASSEL (v. 4.0) pipeline (Bradbury et al. 2007) and an orangethroat darter (*Etheostoma spectabile*) reference genome. We used Bowtie 2.0 for sequence alignment (Langmead & Salzberg 2012). Following alignment, we again removed individuals with low read counts (<300 000). We then used vcftools (Danecek et al. 2011) to filter raw TASSEL-generated SNP datasets. Prior to final SNP filtering, we removed individuals with >35% missing data following preliminary filter exploration (Table S1 in the Supplement at www.int-res.com/articles/suppl/n056p159_supp.pdf). We created final SNP datasets for analysis using 2 sets of filters. For most analyses, we restricted loci to those with 95% data presence, read depths between 20 and 1000, removal of insertions and deletions, loci restricted to a minimum of 1000 base pairs apart, biallelic loci, rare allele frequency of ⩾2%, and loci which passed Hardy-Weinberg equilibrium screening at $\alpha = 0.001$ (Danecek et al. 2011). We also filtered individuals to remove those with >45% missing data, heterozygosity <10%, or mean read depths <25. For analysis of N_e via the linkage-disequilibrium method, we ran similar filters but with rare allele and base pair distance restrictions removed.

2.3. Analyses

We analyzed data at different spatial scales for different groups of analyses. We grouped sites into 5 geographic regions with ⩾19 individuals each (hereinafter: N_e regions, Table 1): Lower Bayou Pierre (LBP), Mainstem Bayou Pierre (MBP), Upper Bayou Pierre (UBP), Bayou Pierre Headwaters (BPH), and White Oak Creek (WOC) for N_e and heterozygosity, due to low numbers of individuals at sites. For population structuring analyses, we analyzed data at the individual level. For all other analyses, we analyzed data at the site level.

We calculated F_{IS}, mean pairwise F_{ST}, and mean pairwise Nei's *D* at the site level using the R package adegenet (Jombart 2008). We tested for significantly positive or negative F_{IS} among sites using a 1-sample *t*-test. We calculated individual heterozygosity as the proportion of heterozygous loci in each individual. Differences in heterozygosity were analyzed using 1-way ANOVA, with Tukey HSD post hoc tests. For both *t*-tests and ANOVA, we used Shapiro-Wilk's tests to assess for normally distributed data, and Bartlett's test for homogeneity of variances. As all data passed these tests (all $p > 0.05$), we proceeded with parametric analyses. We tested for signals of isolation by distance with Nei's *D* at the site level using Mantel's test from the R package vegan (Oksanen et al. 2020), with 10 000 permutations. River distances (km) between sites were calculated by the R package riverdist (Tyers 2020) using the NHDPlus V2 dataset (Moore & Dewald 2016) for isolation by distance analyses. We used the program NeEstimator V2 (Do et

al. 2014) to estimate N_e. We employed the linkage-disequilibrium model recommended by Waples (2016), excluding alleles with a frequency <2%.

We analyzed population structuring using STRUCTURE software v. 2.3.4 (Pritchard et al. 2000). We ran STRUCTURE using K (number of groups) = 1 to 6, with 10 replicates at each level, 750 000 Markov chain repetitions, with a burn-in of 350 000 steps. STRUCTURE was run with the admixture model, with correlated allele frequencies, and no priors based on spatial location. We evaluated the ideal K using the Evanno method (Evanno et al. 2005) and visualized Q scores through the R package pophelper (Francis 2017; our Fig. S1). We further evaluated population structuring and genomic variation using principal coordinates analysis (PCOA). We conducted PCOA using Nei's D and the R function dudi.pco from the package ade4 (Dray & Dufour 2007).

3. RESULTS

Sequence alignment with the *Etheostoma spectabile* genome was modest (~40%). Prior to filtering, we recovered 13 075 SNPs among 135 individuals. After filtering, we retained 845 loci and 114 individuals for most analyses, and 1969 loci and 114 individuals for N_e analysis. No individuals from site 4 passed quality filters; however, raw data are available on Dryad (see 'Data availability'). Numbers of SNPs which failed quality filters are given in Table S2. Individual heterozygosity varied modestly across sample localities (mean = 0.205, range: 0.164–0.249, Fig. S2). Spatially, mean heterozygosity was significantly lower in WOC than in other N_e regions (ANOVA, $F_{4,109}$ = 8.29, p < 0.001, Fig. 3). The highest heterozygosity occurred in the MBP, Turkey Creek, and Foster Creek (Fig. S2, sites 1–3, 5–7, 13; Table 1, Fig. 2). F_{IS} at the site level was significantly negative, suggesting an excess of heterozygotes (t = −43.35, df = 17, p < 0.001, mean = −0.23). We detected a signal of isolation by distance using Nei's genetic distances (Mantel's r = 0.41, p < 0.001). Estimates of N_e were higher for BPH (N_e = 438.9, 95% CI: 343.9–605.4) than more downstream reaches (Table 2). Estimates of N_e for WOC were especially low (N_e = 169.1, 95% CI: 155.3–185.4, Table 2).

Although the ΔK analysis supported the recognition of K = 3 (Fig. 4, Fig. S1), we elected to present the results for both K = 2 and K = 3. At K = 2, group 2 (red, Fig. 4) included individuals with a high degree of ancestry in this group (mean Q = 0.88, SD = 0.14) found across all the sites, although the BPH sites demonstrated the highest Q scores for group 2 (mean Q = 0.95, SD = 0.10). Individuals with group 1 ancestry (dark blue, Fig. 4) were also widely distributed across most sites, but the mean Q scores for this group were lower (mean Q = 0.73, SD = 0.13). At K = 3, the third group (yellow, Fig. 4) was represented by individuals at the BPH sites with high Q scores for this group (mean Q = 0.91, SD = 0.12). Group 2 individuals were still widely distributed across the remaining sites (mean Q = 0.84, SD = 0.18). However, individuals with group 1 ancestry (dark blue, Fig. 4) now mostly tended to show more admixture with group 2 (mean Q = 0.64, SD = 0.15) than was seen with K = 2.

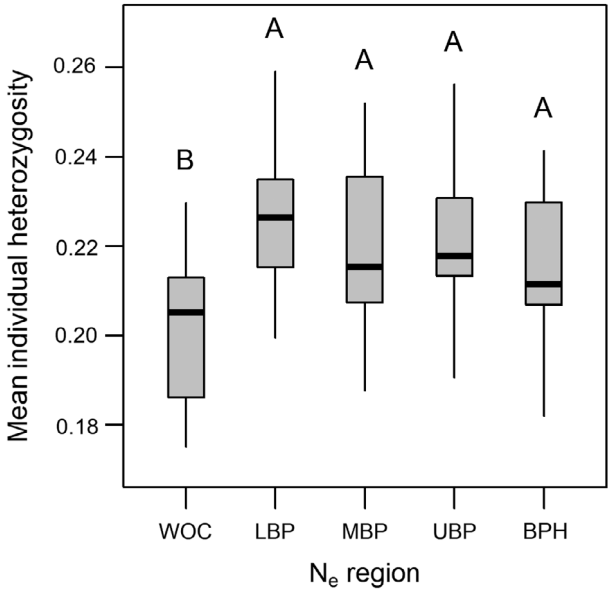

Fig. 3. Individual mean heterozygosity for *Nothonotus rubrus* (across all filtered loci) by effective population size (N_e) region (see Table 1 for abbreviations). Letters indicate results of Tukey HSD tests; shared letters indicate no significant difference between groups. Thick horizontal line: median; box: 25–75% interquartile values; and whiskers: the range of the data

Table 2. Estimates of effective population size (N_e) for *Nothonotus rubrus* across the 5 geographic subdivisions of the Bayou Pierre (see Table 1 for abbreviations), derived from the linkage-disequilibrium method. Estimates of N_e are reported for rare allele frequency cutoff at 0.02 along with the upper and lower limits of the parametric 95% CI. N: number of individuals sequenced

Region	N	N_e estimate	95% Parametric CI Lower	Upper
WOC	25	169.1	155.3	185.4
LBP	28	218.3	199.8	240.5
MBP	21	330.9	284.1	395.5
UBP	21	288.7	251.8	337.9
BPH	19	438.9	343.9	605.4

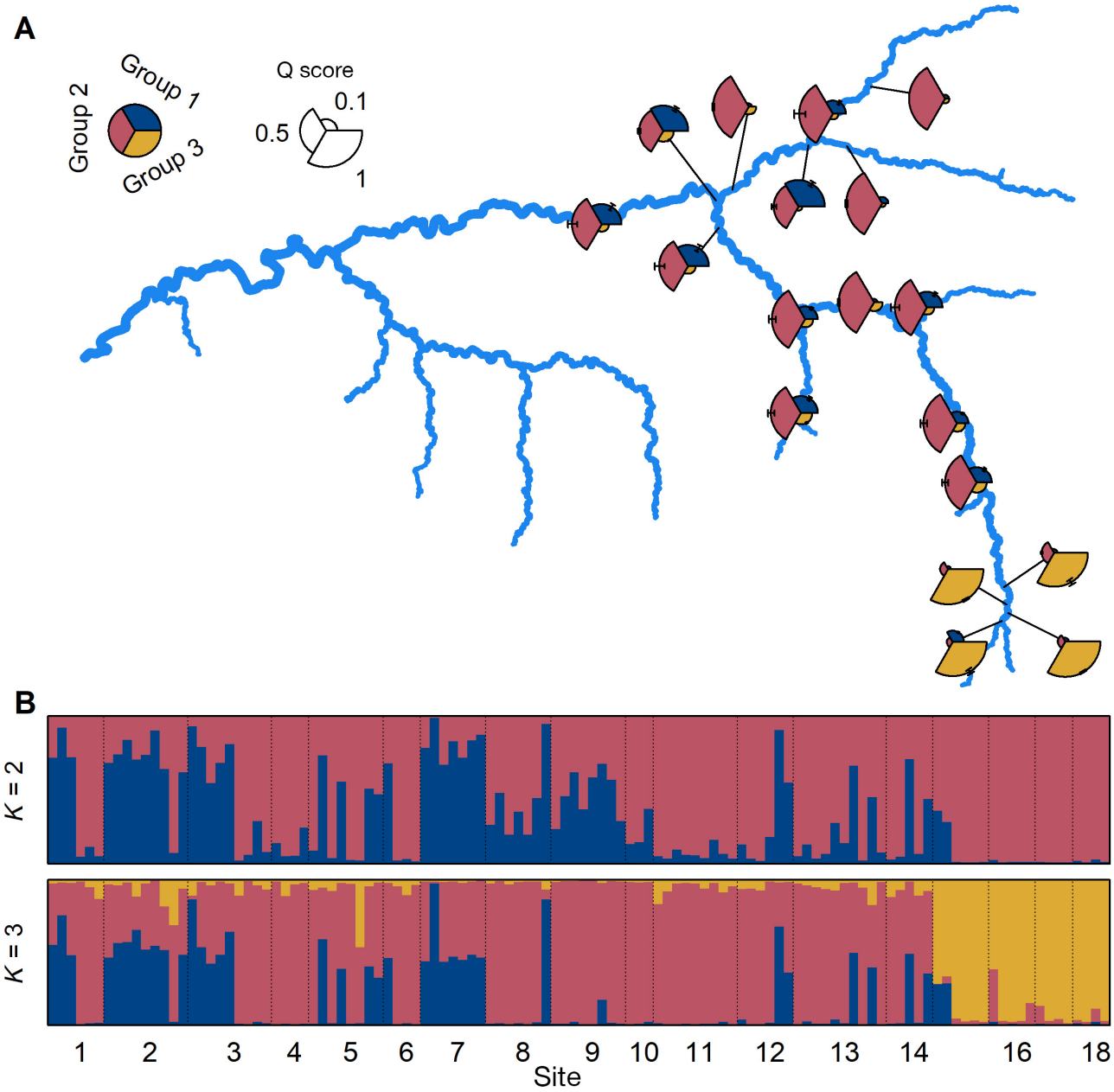

Fig. 4. STRUCTURE plots for *Nothonotus rubrus*. (A) Mean proportion of Q scores for each collection locality. Plots are coxcomb plots, and the distance from center of each wedge indicates the proportion of ancestry for each group. Error bars indicate ± 1 SD. (B) STRUCTURE bar plots indicating proportion of group ancestry within each individual for $K = 2$ (top) and $K = 3$ (bottom). Individuals are in the same order in each plot. Site numbers match Table 1 and Fig. 2

Comparisons of pairwise F_{ST} values at the site scale revealed that the upstream-most locality on Tallahalla Creek (site 11, Table 1, Fig. 2) had a high mean pairwise F_{ST} with all other localities (0.121, range: 0.100–0.146), and that sites in the headwaters of Bayou Pierre were similarly distinct from the remainder of the watershed (0.095, range: 0.066–0.146, Fig. 5). PCOA of individuals suggested substantial genetic structuring, visible on PCOA1 (4.2% of variance) and PCOA2 (3.0% of variance, Fig. 6). Most individuals from the extreme headwater localities scored low on PCOA2, forming a single cluster. Two other clusters of individuals are also apparent, separated on PCOA1. However, in several cases, sites had individuals in both clusters, suggesting multiple ancestry within these localities (Fig. 6). These results are largely concordant with our results from STRUCTURE analysis, and suggest 3 different genetic clusters, with considerable spatial co-occurrence and admixture of 2 of the 3 clusters.

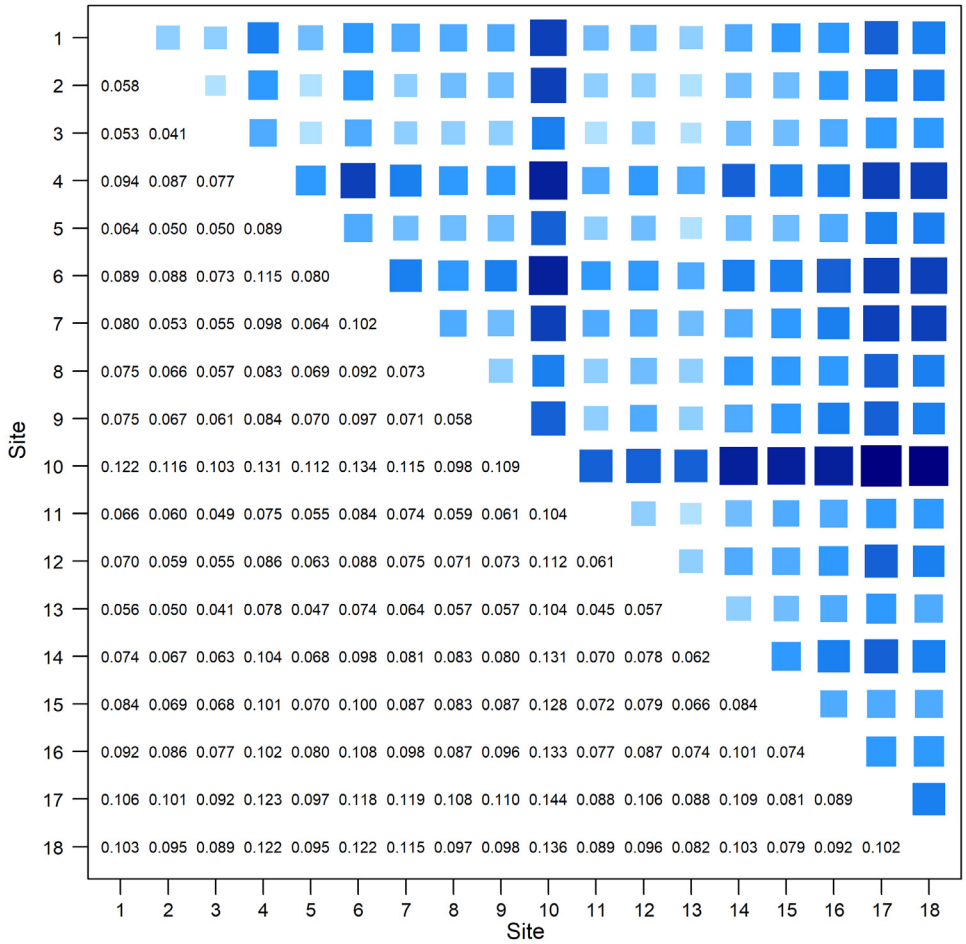

Fig. 5. Pairwise F_{ST} across sample sites for *Nothonotus rubrus*. Square size and darkness indicate F_{ST} value. F_{ST} values in the lower triangle correspond to transposed square size in the upper triangle

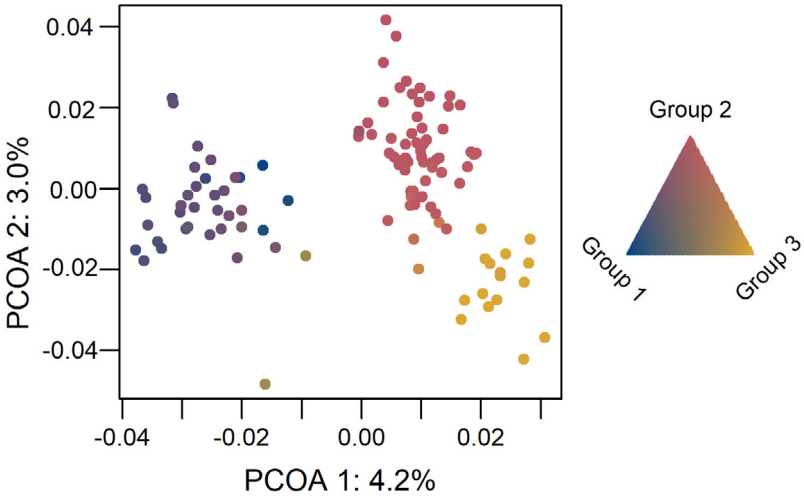

Fig. 6. Principal coordinates analysis (PCOA) of individual Nei's *D* for *Nothonotus rubrus*. Colored points indicate mean Q score for each group from STRUCTURE analysis. Ternary plot in key shows color scaling for different relative proportional ancestry in an individual from each group. Percent variance contained in each component is indicated along each axis

4. DISCUSSION

Our most striking findings were both marked genetic structure across the watershed, but limited structuring among different tributaries. This is in sharp contrast to both previous studies of *Nothonotus rubrus* and of some other darter species in tributaries of the eastern Mississippi Embayment. Previous work using allozymes by Wood (1996) and mitochondrial control region sequences by Slack et al. (2010) found little evidence of genetic structuring in *N. rubrus*. Our high-resolution next-generation sequencing approach provides considerable improvement on past efforts with lower-resolution markers. The relict darter *Etheostoma chienense*, endemic to Bayou de Chien,

possessed extremely limited haplotype diversity and essentially no population structuring in an analysis of both mitochondrial and SNP data (Kattawar & Piller 2020). Sterling et al. (2012) found that, while both Yazoo darters *E. raneyi* and Yoknapatawpha darters *E. faulkneri* had strong population structuring and considerable loss of genetic diversity among populations, population structuring matched tributary isolation by anthropogenic channel straightening, a mechanism related to our initial hypothesis. Unlike these examples with either limited genetic structuring, or structuring driven by tributary arrangement, our data suggest a predominantly linear pattern of population structuring in *N. rubrus*, with little differentiation among the WOC, Foster Creek, and much of the UBP itself. The exceptions to this pattern occur in upper WOC (sites 8–10), where estimates of heterozygosity, pairwise F_{ST}, STRUCTURE, and N_e all suggest either recent colonization of a limited genetic stock or a highly variable sink population. Despite existing museum collections records upstream of the confluence of WOC and Tallahalla Creek back to the 1960s, we are unaware of records of *N. rubrus* in this region until after 1990, supporting recent colonization. Our results overall provide mixed support for our initial hypothesis (evidence of structuring between the headwaters of the mainstem and the rest of the system, evidence suggesting potential recent colonization in the WOC system), and hint at more complex mechanisms driving gene flow in *N. rubrus*.

We can propose some alternate hypotheses to explain the general pattern of population structuring we observed in *N. rubrus*. One potential reason for why we did not detect a pattern similar to Sterling et al. (2012), who also studied a highly eroded system, may be the result of differences in stage and character of erosional processes between Bayou Pierre and tributaries of the Yazoo River system. Unlike Bayou Pierre, the lower portions of the Yazoo River were heavily channelized and dredged in an active effort to reduce flooding and improve agricultural land, instantly isolating tributary habitats (Shankman 1996). In contrast, erosional processes on Bayou Pierre likely stem predominantly from stimuli outside the watershed (Albertson & Patrick 1996, Stearman & Heitmuller 2024). These changes have resulted in widely spaced, dynamic, unstable gravel bars, providing at least some riffle habitats in between tributaries (Stearman & Schaefer 2022), and *N. rubrus* continues to be present below the active erosional crest (Hubbell et al. 2021; our Fig. 2). Further progression of erosional activities may eventually isolate the species in tributaries. Alternatively, life history characteristics and habitat preferences play a strong role in population structuring for darters (Fluker et al. 2014b). *E. raneyi* and *E. faulkneri* belong to the subgenus *Ulocentra* (snubnose darters), which are known for traits related to low dispersal, such as a lack of larval drift (Ruble et al. 2019). Similarly, both *E. raneyi* and *E. faulkneri* prefer small headwater streams (Hubbell et al. 2020), so genetic structuring in these species may predate channelization. *N. rubrus* exhibit larval drift (Slack et al. 2004), indicating higher dispersal and the potential for continued gene flow among tributaries so long as viable habitat patches remain downstream. Finally, patchy or unstable habitats can drive evolution of adaptations for plasticity or dispersal in habitat specialists (Cheptou et al. 2017). Bayou Pierre is believed to have been geomorphically dynamic since at least the beginning of the Holocene in response to channel migration and valley building of the Mississippi River (Stearman & Heitmuller 2024), providing ample opportunity for selective pressure. Other darters with minimal evidence of small-scale population structuring for which this mechanism has been proposed include eastern sand darter *Ammocrypta pellucida*, tuxedo darter *E. lemniscatum*, and tessellated darter *E. olmstedi* (Ginson et al. 2015, Euclide & Marsden 2018, Washburn et al. 2020).

Erosional processes in the watershed still do offer a hypothesis to explain the presence of group 3 within the headwaters. This population occurred entirely above the main knickpoint on the MBP, an erosional feature which has been in existence since at least 1959 and likely decades to centuries earlier (Stearman & Heitmuller 2024). Despite sampling in this portion of the watershed extending back to at least 1968, the first recorded collections of *N. rubrus* in this region occurred in 1988, and records indicate they were uncommon until recently. Plausibly, this population has remained upstream of the knickpoint and has migrated in concert with its movement. As this population was not resolved at $K = 2$, we can infer that this likely occurred after groups 1 and 2 split. We observed *N. rubrus* to be quite abundant at site 14 below the knickpoint, but apart from our own recent collection at site 15, we can find no records of them occurring in the main region of bank failure and channel widening. We propose that the knickpoint itself may be the mechanism separating group 3 from the other groups. The region of active bank failure offers poor habitats for any riffle-dwelling darter species, while regions upstream and downstream offer distinctly different habitats (Fig. 1). With a generation time of 1–2 yr (Knight & Ross 1992), it has been at a minimum 31–62 generations since knickpoint forma-

tion (1959–2021), providing ample time for genetic differentiation. Similar differentiation after relatively short periods of fragmentation has also been observed in numerous other darters, including greenside darter *E. blennioides*, Yazoo darter *E. raneyi*, Kentucky arrow darter *E. spilotum*, Tallapoosa darter *E. tallapoosae*, and logperch *Percina caprodes* (Haponski et al. 2007, Beneteau et al. 2009, Fluker et al. 2014a, Blanton et al. 2019).

Groups 1 and 2 present a more difficult pattern to explain. These 2 populations are co-occurring in contemporary samples, with evidence of considerable admixture, suggesting recent contact. Similar patterns of admixture are readily apparent in recent introgression events of hatchery and wild fishes (Lamaze et al. 2012). We propose 3 (not mutually exclusive) hypotheses for this pattern: a breakdown in some barrier between populations 1 and 2, a loss of downstream habitat for population 1, or a strong selective gradient from upstream to downstream driving differentiation. Considering the first hypothesis, we are unaware of any barrier in the system during the historical period that would have prevented contact between these 2 populations. It is possible that group 1 may have originated in a tributary such as WOC or Little Bayou Pierre, and recently come into contact with group 2 as migrations of the Mississippi River changed the watershed's erosional regime (Stearman & Heitmuller 2024). Second, Stearman & Schaefer (2022) and Stearman & Heitmuller (2024) documented a shift in the lower mainstem from stable channels with abundant bars and riffle–pool structure to wide, unstable channels with a few very large gravel bars and a paucity of riffle–pool structure. Plausibly, group 1 may have originally been downstream and pushed into contact with group 2 due to habitat loss. Our data do not truly test the pattern of the third hypothesis (selective gradient), as such a gradient would be expected to act predominantly on SNPs on regions of the genome undergoing selection (Xu et al. 2017).

Our results carry important conservation implications for managing populations of *N. rubrus*. Unfortunately, genetic diversity is often overlooked in management and conservation planning and actions (Laikre 2010). Estimates of population health that rely solely on numerical abundance or distributions (e.g. Hubbell et al. 2021) do not capture the genetic diversity, which our data revealed. Although our estimates of N_e carry substantial uncertainty, spatial trends are congruent with the findings of Hubbell et al. (2021). Those authors found that relative abundances of *N. rubrus* were highest at the upstream-most end of their distribution, and that downstream populations consisted of few young-of-year individuals (Hubbell et al. 2021). Our estimates of N_e for each region were broadly comparable to other similar range-restricted darters using comparable SNP datasets and methods. Kattawar & Piller (2020) estimated N_e for *E. chienense* at 143.3–448.3, and for Clarks darter *E.* cf. *oophylax* at 154.7–517.3. Fast et al. (2024) estimated N_e for trispot darter *E. trisella* at 342–350, summed across 3 different sampled populations. Our estimates of N_e may be affected by the multiple ancestry in regions where substantial introgression between groups 1 and 2 has occurred. Regardless, the low numbers comparable to other threatened and restricted range darters highlight the importance of conservation efforts for this species.

The introgression of group 1 into group 2 may have conservation implications, which abundance estimates alone would not capture. Genetic introgression among separate populations is not necessarily detrimental to a population, as gene flow can provide beneficial alleles to a population and boost a portfolio effect, buffering populations from environmental stressors (Hamilton & Miller 2016). Indeed, gene flow facilitation may be merited to reduce extinction risks (Pavlova et al. 2017). In contrast to our findings with N_e, *N. rubrus* heterozygosity as inferred from SNPs is closer to values observed for more widespread darter species (e.g. rainbow darter *E. caeruleum*, Luiken et al. 2021) than for other small-range or recently fragmented range headwater darters (e.g. *E. chienense*, bluemask darter *E. akatulo*, Kattawar & Piller 2020, Taylor et al. 2021). However, if local adaptation is already particularly strong, then intermediate phenotypes may suffer a fitness consequence from admixture (Miller et al. 2004, Westley et al. 2013, Fraser et al. 2014). If intermediate phenotypes suffer a sufficient fitness cost but the rate of introgression is sufficiently low, then natural selection may remove an introgressing population (Matthews et al. 2016). Were the second scenario to occur with *N. rubrus*, then either group 1 or group 2's genetic distinctiveness could eventually be lost. As these scenarios depend on the degree to which local adaptation has occurred, we cannot understand the consequences of introgression of populations 1 and 2 until we have resolved whether our observed population structuring also reflects local adaptation to unique habitats.

We recommend special conservation focus for groups 1 and 3, albeit for different reasons. The upstream-most group 3 is currently restricted to approximately 6 km of stream habitat upstream of the knickpoint, although we observed it to be abundant

where it occurs. Stearman & Heitmuller (2024) estimated the knickpoint's upstream migration rate at 216.7 m yr^{-1}. It is likely that the habitat of this population will be altered within 50 yr. Extremely restricted distributions convey a particular risk of extinction from chance events or stochastic fluctuations, as there is no population redundancy or resiliency through rescue effects as would be typical in a broader metapopulation (Staude et al. 2020, Le Feuvre et al. 2021, Dubos et al. 2022). Translocation efforts to Little Bayou Pierre, where the species is absent but similar habitats exist, have been recently suggested as a conservation measure to establish population redundancy for the species as a whole. Translocation efforts are a widespread tool to increase population resiliency and rescue effects, particularly for range-restricted species (George et al. 2009, Pavlova et al. 2017). However, these projects largely have a history of failure and complications, and should be cautiously considered (Berger-Tal et al. 2020). Because the distribution of this group is well understood and so limited, and there is strong evidence of differentiation from other groups, we believe that group 3 should be considered the best potential candidate for any such efforts, should translocation be decided to be prudent. Conversely, the rare downstream group 1 is apparently spatially dispersed, but it is represented in our data by only a handful of relatively pure individuals. We advise that further sampling in the LBP may be necessary to resolve whether any localities exist where this form is not at risk of admixture.

For all populations, we strongly recommend habitat augmentation or restoration efforts which work with, not against, fluvial geomorphic processes. Stream bank stabilization efforts are often the de facto tool of choice to mitigate against headcutting (Bigham 2020). Efforts at stabilization for fish habitat conservation have typically proven effective in the short term (but have been failures in the long term) in the Gulf Coastal Plain (Shields et al. 1994, 1998, 2003, 2007). Given the dire circumstances faced by the habitat of group 3, bank stabilization of the knickpoint may be desirable, but we urge consideration of processes and materials shaping habitats downstream for groups 1 and 2. Stearman & Schaefer (2023) found much of the lower historic range of *N. rubrus* to consist of greatly expanded channels with limited landform diversity, pointing to a widespread scour of materials and reduced landform formation and maintenance. Landform and thus habitat morphogenesis (creation, stability, and evolution) are largely governed by interactions between hydrologic, vegetation, and sedimentary processes and patterns at multiple scales (Wohl et al. 2015, Fryirs 2024, Gurnell & Bertoldi 2024). Of these, direct manipulation of sediment regimes is probably least explored and most promising for Bayou Pierre. Local-scale sediment regime manipulation, through augmentation or redistribution, is rather feasible and has gained widespread acceptance in recent years as a habitat mediation tool (Wohl et al. 2015, Mörtl & De Cesare 2021). This strategy holds promise to help streams rapidly regain landforms fostering diverse aquatic habitats well downstream of the point of intervention (Mörtl & De Cesare 2021). These approaches are not without risk, and may have unintended consequences if not carefully planned (Wohl 2015), but in general they have been extremely successful (Mörtl & De Cesare 2021) and we believe they should at least be considered for habitat restoration or augmentation efforts in the more eroded, low landform diversity reaches.

Our analysis does have its limitations, which affect our ability to draw specific conclusions or interpretations from the data. Notably, our sample size after extraction and sequencing was as low as 3–4 individuals at some localities (Table 1). While robust inferences regarding ancestry can still be drawn from very small numbers of individuals with high numbers of markers (Nazareno et al. 2017, McLaughlin & Winker 2020), this number is on the low end of any published recommendations. Because of the high density of sites in our analysis, we still retain the ability to draw broader conclusions about processes at scales larger than individual sites. Similarly, our relatively modest alignment with our reference genome lowers the number of markers we are able to analyze, although our number of molecular markers still provides orders of magnitude more resolution than previous assessments of *N. rubrus* population structure (Wood 1996, Slack et al. 2010). Even with these limitations, our data still provide an illuminating look into contemporary population structuring in the endemic *N. rubrus*, and valuable insights informing conservation and management goals.

Data availability. All data and code necessary to replicate the analyses in this paper are available at the Dryad data repository (https://doi.org/10.5061/dryad.h70rxwdtm). Full sequence data for all samples are available at SRA GenBank (submission SUB15002021).

Acknowledgements. We acknowledge the substantial help of Matthew Wagner in both design and field components of this project. Langston Haden, Noah Daun, Malia Davidson, Matthew Aiken, Marie Huntsberry, Bailey Coomes, Jonathon Osborne, Robbie Ellwanger, and Calvin Rezac all contributed time in the field for tissue collections. Chris Pellec-

chia conducted DNA extractions and quantifications prior to sequencing. This research was funded by a grant from the US Fish and Wildlife Service (F21AC02855). Analysis and writing were partially supported by a grant from the US Army Corps of Engineers (contract W912HZ21C0064). All fish collection efforts were conducted under an IACUC-approved protocol granted by the University of Southern Mississippi (15102701.1).

LITERATURE CITED

Albertson PE, Patrick DM (1996) Lower Mississippi River tributaries: contributions to the collective science concerning the 'Father of Waters'. Eng Geol 45:383–413

Altermatt F (2013) Diversity in riverine metacommunities: a network perspective. Aquat Ecol 47:365–377

Beneteau CL, Mandrak NE, Heath DD (2009) The effects of river barriers and range expansion of the population genetic structure and stability in greenside darter (*Etheostoma blennioides*) populations. Conserv Genet 10:477–487

Berger-Tal O, Blumstein DT, Swaisgood RR (2020) Conservation translocations: a review of common difficulties and promising directions. Anim Conserv 23:121–131

Bigham KA (2020) Streambank stabilization design, research, and monitoring: the current state and future needs. Trans ASABE 63:351–387

Blanton RE, Cashner MF, Thomas MR, Brandt SL, Floyd MA (2019) Increased habitat fragmentation leads to isolation among and low genetic diversity within populations of the imperiled Kentucky arrow darter (*Etheostoma sagitta spilotum*). Conserv Genet 20:1009–1022

Bradbury PJ, Zhang Z, Kroon DE, Casstevens TM, Ramdoss Y, Buckler ES (2007) TASSEL: software for association mapping of complex traits in diverse samples. Bioinformatics 23:2633–2635

Brauer CJ, Beheregaray LB (2020) Recent and rapid anthropogenic habitat fragmentation increases extinction risk for freshwater biodiversity. Evol Appl 13:2857–2869

Carlson SM, Satterthwaite WH (2011) Weakened portfolio effect in a collapsed salmon population complex. Can J Fish Aquat Sci 68:1579–1589

Carson EW, De la Maza-Benignos M, de Lourdes Lozano-Vilano M, Vela-Valladares L, Banda-Villanueva I, Turner TF (2014) Conservation genetic assessment of the critically endangered Julimes pupfish, *Cyprinodon julimes.* Conserv Genet 15:483–488

Cheptou PO, Hargreaves AL, Bonte D, Jacquemyn H (2017) Adaptation to fragmentation: evolutionary dynamics driven by human influences. Philos Trans R Soc B 372: 20160037

Clemens BJ, Wang CJ (2021) Dispelling misperceptions of native lampreys (*Entosphenus* and *Lampetra* spp.) in the Pacific Northwest (USA). Conserv Sci Pract 3:e402

Cluer B, Thorne C (2014) A stream evolution model integrating habitat and ecosystem benefits. River Res Appl 30: 135–154

Coleman RA, Gauffre B, Pavlova A, Beheregaray LB and others (2018) Artificial barriers prevent genetic recovery of small isolated populations of a low-mobility freshwater fish. Heredity 120:515–532

Danecek P, Auton A, Abecasis G, Albers CA and others (2011) The variant call format and VCFtools. Bioinformatics 27:2156–2158

Do C, Waples RS, Peel D, Macbeth GM, Tillett BJ, Ovenden JR (2014) NeEstimator v2: re-implementation of software for the estimation of contemporary effective population size (N_e) from genetic data. Mol Ecol Resour 14:209–214

Dray S, Dufour AB (2007) The ade4 package: implementing the duality diagram for ecologists. J Stat Softw 22:1–20

Dubos N, Montfort F, Grinand C, Nourtier M and others (2022) Are narrow-ranging species doomed to extinction? Projected dramatic decline in future climate suitability of two highly threatened species. Perspect Ecol Conserv 20:18–28

Elshire RJ, Glaubitz JC, Sun Q, Poland JA, Kawamoto K, Buckler ES, Mitchell SE (2011) A robust, simple genotype-by-sequencing (GBS) approach for high diversity species. PLOS ONE 6:e19379

Euclide P, Marsden JE (2018) Role of drainage and barriers in the genetic structuring of a tessellated darter population. Conserv Genet 19:1379–1392

Evanno G, Regnaut S, Goudet J (2005) Detecting the number of clusters of individuals using the software STRUCTURE: a simulation study. Mol Ecol 14:2611–2620

Fagan WF (2002) Connectivity, fragmentation, and extinction risk in dendritic metapopulations. Ecology 83: 3243–3249

Fast KM, Fluker BL, Kuhajda BR, O'Neil PE, McGregor SW, Piteo MS, Sandel MW (2024) Conservation genomics of the threatened trispot darter (*Etheostoma trisella*). Conserv Genet 25:291–304

Faulks LK, Gilligan DM, Beheregaray LB (2011) The role of anthropogenic vs. natural in-stream structures in determining connectivity and genetic diversity in an endangered freshwater fish, Macquarie perch (*Macquaria australasica*). Evol Appl 4:589–601

Fluker BL, Kuhajda BR, Harris PM (2014a) The effects of riverine impoundment on genetic structure and gene flow in two stream fishes in the Mobile River basin. Freshw Biol 59:526–543

Fluker BL, Kuhajda BR, Harris PM (2014b) The influence of life-history strategy on genetic differentiation and lineage divergence in darters (Percidae: Etheostomatinae). Evolution 68:3199–3216

Francis RM (2017) Pophelper: an R package and web app to analyse and visualize population structure. Mol Ecol Resour 17:27–32

Fraser DJ, Debes PV, Bernatchez L, Hutchings JA (2014) Population size, habitat fragmentation, and the nature of adaptive variation in a stream fish. Proc R Soc B 281: 20140370

Fryirs K (2024) Leveraging understandings of biogeomorphic river recovery to reframe river management philosophy and communication strategies. River Res Appl 40:1142–1150

Fuller MR, Doyle MW, Strayer DL (2015) Causes and consequences of habitat fragmentation in river networks. Ann NY Acad Sci 1355:31–51

George AL, Kuhajda BR, Williams JD, Cantrell MA, Rakes PL, Shute JR (2009) Guidelines for propagation and translocation for freshwater fish conservation. Fisheries 34: 529–545

Ginson R, Walter RP, Mandrak NE, Beneteau CL, Heath DD (2015) Hierarchical analysis of genetic structure in the habitat-specialist eastern sand darter (*Ammocrypta pellucida*). Ecol Evol 5:695–708

Gran KB, Czuba JA (2017) Sediment pulse evolution and the role of network structure. Geomorphology 277:17–30

Gurnell AM, Bertoldi W (2024) Plants and river morpho-

dynamics: the emergence of fluvial biogeomorphology. River Res Appl 40:887–942
Hamilton JA, Miller JM (2016) Adaptive introgression as a resource for management and genetic conservation in a changing climate. Conserv Biol 30:33–41
Haponski AE, Marth TA, Stepien CA (2007) Genetic divergence across a low-head dam: a preliminary analysis using logperch and greenside darters. J Gt Lakes Res 33(Suppl 2):117–126
Horton JD, San Juan CA, Stoeser DB (2017) The State Geologic Map Compilation (SGMC) geodatabase of the conterminous United States, 1.1. US Geological Survey Data Series 1052, US Geological Survey, Reston, VA
Hubbell JP, Schaefer JF, Warren ML Jr, Sterling KA (2020) Modelling patterns of coexistence of three congeneric headwater fishes. Freshw Biol 65:1017–1027
Hubbell J, Stearman LW, Schaefer JF (2021) Updated status of Bayou darter, a narrowly ranged endemic in a geomorphically active watershed. Endang Species Res 44: 137–148
Jombart T (2008) adegenet: a R package for the multivariate analysis of genetic markers. Bioinformatics 24:1403–1405
Kattawar J, Piller KR (2020) Comparative population genetics of the federally endangered relict darter, and its sister taxon the Clarks darter (Teleostei: Percidae). Conserv Genet 21:957–970
Keefer ML, Caudill CC, Clabough TS, Jepson MA and others (2013) Fishway passage bottleneck identification and prioritization: a case study of Pacific lamprey at Bonneville Dam. Can J Fish Aquat Sci 70:1551–1565
Knight JG, Ross ST (1992) Reproduction, age and growth of the Bayou darter *Etheostoma rubrum* (Pisces, Percidae): an endemic of Bayou Pierre. Am Midl Nat 127:91–105
Laikre L (2010) Genetic diversity is overlooked in international conservation policy implementation. Conserv Genet 11:349–354
Lamaze FC, Sauvage C, Marie A, Garant D, Bernatchez L (2012) Dynamics of introgressive hybridization assessed by SNP population genomics of coding genes in stocked brook charr (*Salvelinus fontinalis*). Mol Ecol 21:2877–2895
Langmead B, Salzberg SL (2012) Fast gapped-read alignment with Bowtie 2. Nat Methods 9:357–359
Le Feuvre MC, Dempster T, Shelley JJ, Davis AM, Swearer SE (2021) Range restriction leads to narrower ecological niches and greater extinction risk in Australian freshwater fish. Biodivers Conserv 30:2955–2976
Lofthouse C, Robert A (2008) Riffle-pool sequences and meander morphology. Geomorphology 99:214–223
Luiken JM, Gamble T, Berendzen PB (2021) Comparative riverscape genomics of the rainbow darter (*Etheostoma caeruleum*) in glaciated and unglaciated environments. Ecol Evol 11:18305–18318
Matthews WH (1978) Fishes of Bayou Pierre, southwest Mississippi. Master's thesis, Northeast Louisiana University, Monroe, LA
Matthews WJ, Turner TF, Osborne MJ (2016) Breakdown of a hybrid swarm between two darters (Percidae), *Etheostoma radiosum* and *Etheostoma spectabile*, with loss of one parental species. Copeia 104:873–878
McLaughlin JF, Winker K (2020) An empirical examination of sample size effects on population demographic estimates in birds using single nucleotide polymorphism (SNP) data. PeerJ 8:e9939
Miller LM, Close T, Kapuscinski AR (2004) Lower fitness of hatchery and hybrid rainbow trout compared to naturalized populations in Lake Superior tributaries. Mol Ecol 13:3379–3388
Moore RB, Dewald TG (2016) The road to NHDPlus — advancements in digital stream networks and associated catchments. J Am Water Resour Assoc 52:890–900
Mörtl C, De Cesare G (2021) Sediment augmentation for river rehabilitation and management — a review. Land (Basel) 10:1309
Murray KD, Borevitz JO (2018) Axe: rapid, competitive sequence read demultiplexing using a trie. Bioinformatics 34:3924–3925
Nazareno AG, Bemmels JB, Dick CW, Lohmann LG (2017) Minimum sample sizes for population genomics: an empirical study from an Amazonian plant species. Mol Ecol Resour 17:1136–1147
Oksanen J, Blanchet FG, Friendly M, Kindt R and others (2020) vegan: community ecology package. https://CRAN.R-project.org/package=vegan
Pavlova A, Beheregaray LB, Coleman R, Gilligan D and others (2017) Severe consequences of habitat fragmentation on genetic diversity of an endangered Australian freshwater fish: a call for assisted gene flow. Evol Appl 10: 531–550
Perkin JS, Gido KB (2011) Stream fragmentation thresholds for a reproductive guild of Great Plains fishes. Fisheries 36:371–383
Perkin JS, Gido KB (2012) Fragmentation alters stream fish community structure in dendritic ecological networks. Ecol Appl 22:2176–2187
Pritchard JK, Stephens M, Donnelly P (2000) Inference of population structure using multilocus genotype data. Genetics 155:945–959
Raney EC, Suttkus RD (1966) *Etheostoma rubrum*, a new percid fish of the subgenus *Nothonotus* from Bayou Pierre, Mississippi. Tulane Stud Zool 13:95–102
Ross ST, Knight JG, Wilkins SD (1992) Distribution and microhabitat dynamics of the threatened Bayou darter, *Etheostoma rubrum.* Copeia 1992:658–671
Ross ST, O'Connell MT, Patrick DM, Latorre CA, Slack WT, Knight JG, Wilkins SD (2001) Stream erosion and densities of *Etheostoma rubrum* (Percidae) and associated riffle-inhabiting fishes: biotic stability in a variable habitat. Copeia 2001:916–927
Ruble CL, Sterling KA, Warren ML (2019) Captive propagation and early life history of the Yazoo darter (*Etheostoma raneyi*). Southeast Nat 18:525–540
Shankman D (1996) Stream channelization and changing vegetation patterns in the U.S. Coastal Plain. Geogr Rev 86:216–232
Shields FD Jr, Knight SS, Cooper CM (1994) Effects of channel incision on base flow stream habitats and fishes. Environ Manage 18:43–57
Shields FD Jr, Knight SS, Cooper CM (1998) Rehabilitation of aquatic habitats in warmwater streams damaged by channel incision in Mississippi. Hydrobiologia 382: 63–86
Shields FD Jr, Knight SS, Morin N, Blank J (2003) Response of fishes and aquatic habitats to sand-bed stream restoration using large woody debris. Hydrobiologia 494:251–257
Shields FD Jr, Knight SS, Cooper CM (2007) Can warmwater streams be rehabilitated using watershed-scale standard erosion control measures alone? Environ Manage 40: 62–79
Simon A, Rinaldi M (2006) Disturbance, stream incision, and channel evolution: the roles of excess transport capacity

and boundary materials in controlling channel response. Geomorphology 79:361–383
Slack WT, Ross ST, Ewing JA (2004) Ecology and population structure of the Bayou darter, *Etheostoma rubrum*: disjunct riffle habitats and downstream transport of larvae. Environ Biol Fishes 71:151–164
Slack WT, Sumners JA, Rooney AP, Taylor CM (2010) Conservation genetics of the threatened Bayou darter (Percidae: *Etheostoma rubrum*) in the Bayou Pierre system of southwestern Mississippi. Copeia 2010:176–180
Stanford JA, Ward JV, Liss WJ, Frissell CA, Williams RN, Lichatowich JA, Coutant CC (1996) A general protocol for restoration of regulated rivers. Regul Rivers 12:391–413
Staude IR, Navarro LM, Pereira HM (2020) Range size predicts the risk of local extinction from habitat loss. Glob Ecol Biogeogr 29:16–25
Stearman LW (2023) A multiscale linkage between riverscape and fish community coevolution. Dissertation, University of Southern Mississippi, Hattiesburg, MS
Stearman LW, Heitmuller FT (2024) Historical and contemporary drivers of knickpoint retreat and morphological evolution along Bayou Pierre, Mississippi. River Res Appl 40:693–707
Stearman LW, Lynch DT (2013) Patterns of assemblage change in prairie stream fishes in relation to urban stormwater impoundments. Hydrobiologia 718:221–235
Stearman LW, Schaefer JF (2022) Long-term minnow community trait shifts and metacommunity dynamics in a geomorphically unstable river. Ecol Freshw Fish 31: 662–674
Stearman LW, Schaefer JF (2023) Fluvial geomorphic evolution and stream fish community trajectories in the Bayou Pierre, Mississippi. Freshw Biol 68:2011–2026
Stearman LW, Schaefer JF (2024) Altered metapopulation dynamics in a headwater specialist in geomorphically dynamic catchments. Freshw Biol 69:1203–1217
Sterling KA, Reed DH, Noonan BP, Warren ML (2012) Genetic effects of habitat fragmentation and population isolation on *Etheostoma raneyi* (Percidae). Conserv Genet 13:859–872
Suttkus RD, Clemmer GH (1977) A status report on the Bayou darter, *Etheostoma rubrum*, and the Bayou Pierre system. Southeast Fishes Counc Proc 1:1–2
Taylor LU, Benavides E, Simmons JW, Near TJ (2021) Genomic and phenotypic divergence informs translocation strategies for an endangered freshwater fish. Mol Ecol 30:3394–3407
Tipton JA, Bart HL, Piller KR (2004) Geomorphic disturbance and its impact on darter (Teleostomi: Percidae) distribution and abundance in the Pearl River drainage, Mississippi. Hydrobiologia 527:49–61
Tyers M (2020) riverdist: river network distance computation and applications. R package version 0.15.5. https://CRAN.R-project.org/package=riverdist
Valdés González A, Martínez Estévez L, Ángeles Villeda Ma E, Ceballos G (2020) The extinction of the Catarina pupfish *Megupsilon aporus* and the implications for the conservation of freshwater fish in Mexico. Oryx 54:154–160
Van Dyke C (2013) Channels in the making—an appraisal of channel evolution models. Geogr Compass 7:759–777
Waples RS (2016) Making sense of genetic estimates of effective population size. Mol Ecol 25:4689–4691
Washburn BA, Cashner MF, Blanton RE (2020) Small fish, large river: surprisingly minimal genetic structure in a dispersal-limited, habitat specialist fish. Ecol Evol 10: 2253–2268
Westley PAH, Ward EJ, Fleming IA (2013) Fine-scale local adaptation in an invasive freshwater fish has evolved in contemporary time. Proc R Soc B 280:20122327
Weston MR, Johnson RL, Christian AD (2010) Niche partitioning of the sympatric yellowcheek darter *Etheostoma moorei* and rainbow darter *Etheostoma caeruleum* in the Little Red River, Arkansas. Am Midl Nat 164:187–200
Wohl E (2015) Legacy effects on sediments in river corridors. Earth Sci Rev 147:30–53
Wohl E, Bledsoe BP, Jacobson RB, Poff NL, Rathburn SL, Walters DM, Wilcox AC (2015) The natural sediment regime in rivers: broadening the foundation for ecosystem management. Bioscience 65:358–371
Wood RM (1996) Phylogenetic systematics of the darter subgenus *Nothonotus* (Teleostei: Percidae). Copeia 1996: 300–318
Xu S, Song N, Zhao L, Cai S, Han Z, Gao T (2017) Genomic evidence for local adaptation in the ovoviviparous marine fish *Sebastiscus marmoratus* with a background of population homogeneity. Sci Rep 7:1562

Editorial responsibility: Brendan Godley, University of Exeter, Cornwall Campus, UK
Reviewed by: 3 anonymous referees
Submitted: May 24, 2024; Accepted: January 13, 2025
Proofs received from author(s): March 21, 2025

This article is Open Access under the Creative Commons by Attribution (CC-BY) 4.0 License, https://creativecommons.org/licenses/by/4.0/deed.en. Use, distribution and reproduction are unrestricted provided the authors and original publication are credited, and indicate if changes were made